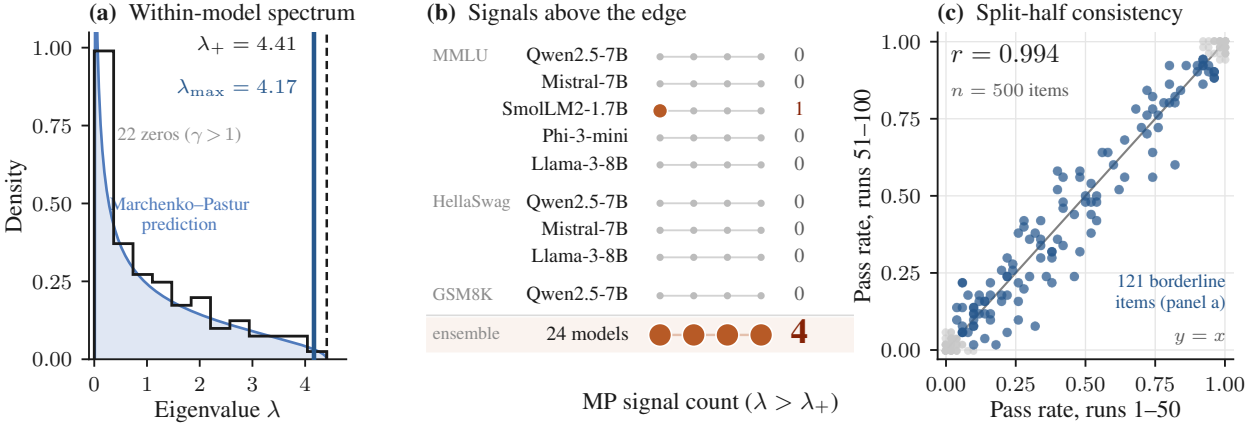

(a) Within-model spectrum
$\lambda_+ = 4.41$
$\lambda_{\max} = 4.17$
22 zeros ($\gamma > 1$)
Marchenko–Pastur prediction
Density
Eigenvalue λ
(b) Signals above the edge
MMLU
Qwen2.5-7B
Mistral-7B
SmolLM2-1.7B
Phi-3-mini
Llama-3-8B
HellaSwag
Qwen2.5-7B
Mistral-7B
Llama-3-8B
GSM8K
Qwen2.5-7B
ensemble
24 models
4
MP signal count ($\lambda > \lambda_+$)
(c) Split-half consistency
$r = 0.994$
$n = 500$ items
121 borderline items (panel a)
$y = x$
Pass rate, runs 51–100
Pass rate, runs 1–50